

Teichmüller space and Fenchel-Nielsen coordinate

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Abstract

In this note, we review basic facts on Teichmüller space and explain the Fenchel-Nielsen coordinate. Our discussion follows two standard textbooks [1, 2]. Throughout this note, we consider only orientable surfaces.

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1 Introduction

It is a blessing to encounter a subject that is simple enough to allow explicit solutions and rich enough to relate various different concepts. The space of two-dimensional surfaces is such an example. Unlike curves, there are infinitely many topologically inequivalent surfaces that are labeled, simply, by an integer, the genus. This is the topological classification of surfaces (often attributed to Möbius)

Theorem 1.1 (Topological classification of surfaces). *Any closed connected oriented surface is homeomorphic to exactly one of the surfaces S_g for $g = 0, 1, 2, \dots$, where S_g is the connected sum of g tori. Any compact, connected, orientable surface is obtained from a closed surface by removing $b \geq 0$ open disks with disjoint closures.*

Additional compatible structures on a surface is strongly constrained. For example, the Gauss-Bonnet theorem shows how the curvature is constrained demonstrated by the topology

$$\int_S R \sqrt{g} d^2x = 2\pi \chi(S), \quad \chi(S) = 2 - 2g - (b + n) \quad (1)$$

where b is the number of removed disks (boundary components), n is the number of punctures, $\chi(S)$ is the Euler characteristic. We want to convey the feeling that there are a variety of two-dimensional surfaces. They have a more rigid structure than higher dimensional objects, which, in some sense, explains the possibility of having a complete classification and a unified descriptions of different structures.

In this note, we focus on one small aspect: how to characterize the (1) complex structure (2) conformal structure and (3) hyperbolic structure of compact surfaces? These three structures, defined differently, are in one-to-one correspondence with each other, which can be captured by a unifying framework. This is known as the Teichmüller theory.¹ This note only discusses the Fenchel-Nielsen coordinate, a coordinate system for the space of inequivalent hyperbolic structures of two-dimensional surfaces, i.e., the Teichmüller space.

Given a surface of genus g , the Teichmüller space, a priori, can be infinite-dimensional. Fortunately, the task at hand is much simpler than what it seems to be. The strategy is to “divide and conquer”. Specifically, we first divide the whole surface into minimal building blocks, which only have one type, the pair of pants. Surprisingly, we will see that the Teichmüller space of a pair of pants is only three-dimensional. Then, we list inequivalent ways to glue these pairs of pants together, which are also characterized by a finite number of parameters. They together give us the Fenchel-Nielsen coordinate and also shows that the Teichmüller space is finite-dimensional.

The rest of the note is organized as follows. Sec. 2 introduces the definition of complex structure, conformal structure and hyperbolic structure, and explain their equivalence. The rest of the note will speak in the language of hyperbolic structure. Sec. 3 defines the space of hyperbolic structures and introduces the pair-of-pants decomposition. Sec. 4 discusses the hyperbolic structure of a single pair of pants, the building block of the whole surface. It also discusses the collar lemma. Sec. 5 discusses the process of gluing two pairs of pants and the associated extra data. Sec. 6 assembles the previous two sections and introduces the Fenchel-Nielsen coordinate. We also introduce a concrete method of calculating these coordinate and provide concrete examples.

¹The origin of Teichmüller theory can be traced back to Riemann’s work on multi-valued complex functions. Teichmüller provided more precise meaning for Riemann work, who defined the space, introduced metric, and equipped it with complex structures. Unfortunately, Teichmüller died young during the World War. His sketch of the theory was left for other mathematicians to develop.

2 Complex = conformal = hyperbolic

In this section, we review the basic definitions of complex structure, conformal structure, hyperbolic structure, and establish their equivalence. In a word, this section gives two equivalent versions of the uniformization theorem.

2.1 Review of definitions

Complex structure Roughly speaking, a complex structure is to describe a surface using complex coordinates so that we can define holomorphic functions. However, there may not exist a globally defined coordinate system. We need to define it patch by patch.

Definition 2.1 (Complex structure). *Let S be a connected Hausdorff space. A (one-dimensional) complex structure on S is a family of coordinate charts $\{(U_i, \phi_i)\}$ that satisfies three conditions*

1. *Each U_i is a open set, and $R = \cup_i U_i$*
2. *Each $\phi_i : U_i \rightarrow \mathbb{C}$ gives a homeomorphism of U_i onto an open subset in the complex plane $\mathbb{C}$. It defines a complex coordinate on U_i .*
3. *On the overlapping region $U_i \cap U_j \neq \emptyset$, the transition function and its inverse are both holomorphic. Namely,*

$$\phi_{ij} = \phi_i \circ \phi_j^{-1} : \phi_j(U_i \cap U_j) \rightarrow \phi_i(U_i \cap U_j) \quad (2)$$

is biholomorphic.

A connecte Hausdorff space with a one-dimensional complex structure is called a Riemann surface.

Multi-valued complex functions provide the classic examples of Riemann surfaces. Take $w = \sqrt{z}$ as an example. First of all, we can view it as a surface $S = \{(z, w) \in \mathbb{C}^2 | z = w^2\}$. To equip it with a complex structure, we can choose a single chart and use $\phi(p) = w \in \mathbb{C}$ as a globally defined complex coordinate. It also shows that S is biholomorphically equivalent to $\mathbb{C}$. We can also treat S as a two-sheeted covering space of the z -plane. Specifically, the function $z = w^2$ gives a biholomorphic map from both the upper and lower half of the w -plane to the slit plane $\mathbb{C} \setminus [0, +\infty)$. We can take two slit planes and glue them along the slit to obtain S . Each slit plane is a sheet, which has the coordinate $\tilde{\phi}(p) = z$ and the inverse map $\tilde{\phi}^{-1}(z) = (z, \pm\sqrt{z})$. Strictly speaking, the slit planes do not cover S , but it is straightforward to add another two open sets to cover S and equip it with a full-fledged complex structure for S . The two ways of defining complex structures are of course equivalent, as we discuss below.

Given a complex structure $\{(U, \phi_i)\}$, a *coordinate neighborhood* (U, ϕ) of a Riemann surface S is defined such that

$$\phi \circ \phi_i^{-1} : \phi_i(U \cap U_i) \rightarrow \phi(U \cap U_i) \quad (3)$$

are biholomorphic on the non-empty overlapping regions. A holomorphic function on S is a function $f : S \rightarrow \mathbb{C}$ such that $f \circ \phi^{-1}$ is holomorphic on each coordinate neighborhood (U, ϕ) . A continuous map $F : R \rightarrow S$ between two Riemann surface is holomorphic if every point $p \in R$ has a coordinate neighborhood (U, ϕ) on R and $F(p)$ has $(V, \tilde{\phi})$ on S such that

$$\tilde{\phi} \circ F \circ \phi^{-1} : \phi(U) \rightarrow \tilde{\phi}(V) \quad (4)$$

is holomorphic.

Definition 2.2 (Biholomorphic equivalence). *Two Riemann surfaces R and S are biholomorphic equivalent if there is a biholomorphic map between them. In this case, we regard R and S as the same Riemann surface, they have the same complex structure.*

Conformal structure Colloquially speaking, a conformal structure does not care about the length but only the angles.

Definition 2.3 (Conformal structure). *Let g_1 and g_2 be two Riemannian metric on the surface S . They are conformally equivalent if $g_1 = e^\sigma g_2$ for some scalar function $\sigma : S \rightarrow \mathbb{R}$. Each conformal equivalence class is called a conformal structure.*

It follows immediately that the conformal structure plays an important role in 2D conformal field theories (CFTs). Specifically, the CFT partition function on a closed surface with the (reference) metric $\hat{g}$ and the conformally equivalent metric $g_{\mu\nu} = e^\sigma \hat{g}_{\mu\nu}$ are related by

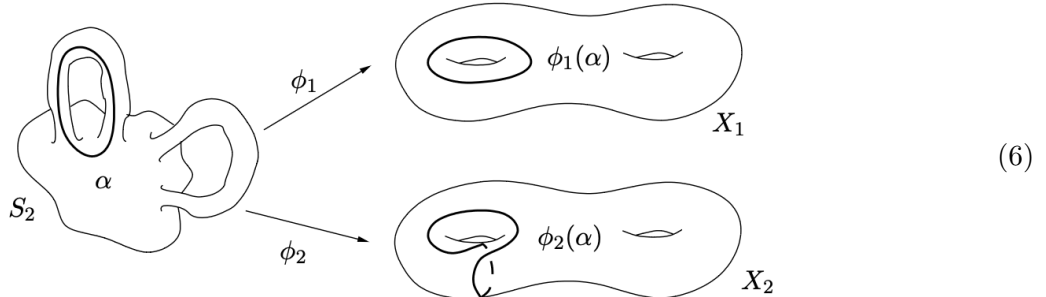
$$Z[g] = e^{\frac{c}{24\pi} S_L} Z[\hat{g}], \quad S_L = \frac{1}{2} \int \left(\hat{R}\sigma + \frac{1}{2} \hat{g}^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma \right) \sqrt{\hat{g}} d^2 x \quad (5)$$

Note that the ratio is independent of any CFT details except for the central charge. Therefore, the non-trivial dependence on the CFT data is entirely through the conformal structure.

Hyperbolic structure A class of particularly simple surfaces comprises the uniformly curved surfaces. By the Gauss-Bonnet theorem, a non-negative constant curvature yields limited choices whereas a negative constant curvature still saves a plenty room (as long as the surface has a negative Euler characteristic). This is what a hyperbolic structure is concerned with. We say that a surface admits a *hyperbolic metric* if there exists a complete, finite-area Riemannian metric on S of constant Gauss curvature -1 , where the geodesics in ∂S are geodesics in S as well.² For simplicity, we will use the term *hyperbolic surface* to denote a surface endowed with a coordinate system and a fixed hyperbolic metric.

Definition 2.4 (Hyperbolic structure). *Let S be a topological surface with $\chi(S) < 0$. A hyperbolic structure is an orientation-preserving homeomorphism $\phi : S \rightarrow X$, where X is a surface with a complete, finite-area hyperbolic metric with totally geodesic boundary. We record the hyperbolic structure by the pair (X, ϕ) , where ϕ is called the marking, and either X or the pair (X, ϕ) is called a marked hyperbolic surface.*

The marking $\phi : S \rightarrow X$ induces a hyperbolic metric on S itself. Therefore, we have the more direct definition: a hyperbolic structure on S is the admitted hyperbolic metric. The more abstract definition—introducing an auxiliary surface X and defining it as the pair (X, ϕ) —has some advantages. It cleanly illustrates the conceptual separation between the geometry (hyperbolic structure) and the topology. It also makes it easier to visualize inequivalent hyperbolic structures. An example is drawn below (figure from Ref. [2]), where the closed curve α is mapped to curves that are not homotopic to each other.



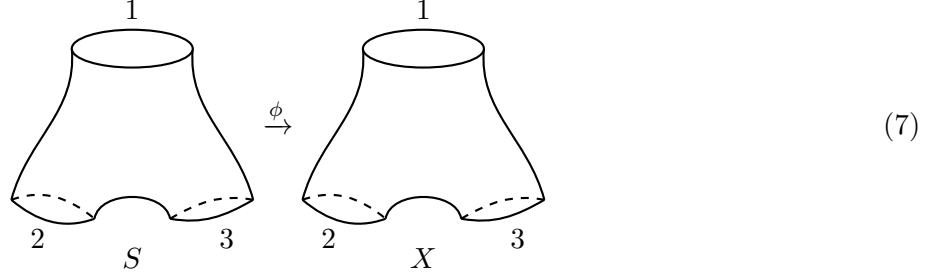
²We note a difference in the definition of Gauss curvature K and scalar curvature $R = 2K$. Here we mean $K = -1$ and thus $R = -2$. For the reader's reference, the Poincaré upper half-plane $ds^2 = (dx^2 + dy^2)/y^2$ has $K = -1$, $R = -2$.

We now give a more precise definition on the equivalence between hyperbolic structures

Definition 2.5. *Two marked hyperbolic structures (X_1, ϕ_1) and (X_2, ϕ_2) are equivalent if there exists an isometry $I : X_1 \rightarrow X_2$ such that $I \circ \phi_1$ and ϕ_2 are homotopic.*

Here X_1 and X_2 are the same geometric object that is endowed with different coordinate systems, and thus the isometry is really a coordinate transformation. We require the isometry I itself is homotopic. This is clear through the direct approach: defining the hyperbolic structure as the hyperbolic metric on S , the equivalence relation is generated by modding out the isometry $I \in \text{Diff}_0(S)$.

The requirement $I \in \text{Diff}_0(S)$ also preserves the marking. We drew an example below, which we hope also explains the name “marking”



Here S is a topological pair of pants and X is a hyperbolic pair of pants with the three boundaries sharing the same geodesic length. The marking ϕ_1 mapping i -th cuff to the i -th cuff and the marking ϕ_2 mapping i -th cuff to $(i + 1)$ -th cuff are the same up to a permutation of the boundary. Since permutation is not homotopic to the identity isometry, ϕ_1 and ϕ_2 are not equivalent. They are regarded as different elements in the Teichmüller space. This is essentially the distinction between the Teichmüller space and the moduli space.

2.2 Equivalence and uniformization theorem

We establish their equivalence by first showing that complex structure is equivalent to conformal structure, and show that conformal structure is equivalent to hyperbolic structure.

Complex = conformal We first show that a conformal structure induces a unique complex structure. In a coordinate chart $(U, (x, y))$, we have

$$ds^2 = E dx^2 + 2F dx dy + G dy^2 \quad (8)$$

Using the complex coordinate $z = x + iy$, we have

$$ds^2 = \lambda |dz + \mu d\bar{z}|^2 \quad (9)$$

where $\lambda > 0$ and μ is complex with the following explicit expressions

$$\lambda = \frac{E + G + 2\sqrt{EG - F^2}}{4}, \quad \mu = \frac{E - G + 2iF}{E + G + 2\sqrt{EG - F^2}} \quad (10)$$

In particular, μ is called the *Beltrami differential* and plays an important role in Teichmüller theory.³ We can then choose a new complex coordinate $w = w(z, \bar{z})$ so that the metric fall into an isothermal

³Under a holomorphic coordinate transformation $z = z(\zeta)$, we have $ds^2 = \lambda' |d\zeta + \mu' d\bar{\zeta}|^2$, where $\lambda' = |dz/d\zeta|^2 \lambda$ and $\mu' = (dz/d\zeta)^{-1} (d\bar{z}/d\bar{\zeta}) \mu$. For this reason, the Beltrami differential is a differential form of weight $(-1, 1)$. Under a general coordinate transformation $z = z(\zeta, \bar{\zeta})$, the Beltrami differential transforms as

$$\mu(z, \bar{z}) \mapsto \mu'(\zeta, \bar{\zeta}) = \frac{(\partial z / \partial \bar{\zeta}) + \mu(\partial \bar{z} / \partial \bar{\zeta})}{(\partial z / \partial \zeta) + \mu(\partial \bar{z} / \partial \zeta)} \quad (11)$$

form

$$ds^2 = \rho dw d\bar{w}, \quad \rho > 0. \quad (12)$$

This new coordinate must satisfy the Beltrami equation

$$\frac{\partial w}{\partial \bar{z}} = \mu \frac{\partial w}{\partial z} \quad (13)$$

Importantly, this equation is independent of λ , and the solution w is a function only of the conformal structure. For this being an orientation-preserving diffeomorphism, we need a positive Jacobian, i.e.,

$$\left| \frac{\partial w}{\partial z} \right|^2 - \left| \frac{\partial w}{\partial \bar{z}} \right|^2 = (1 - |\mu|^2) \left| \frac{\partial w}{\partial z} \right|^2 > 0 \Rightarrow |\mu| < 1 \quad (14)$$

This condition also guarantees that the Beltrami equation has a globally defined solution [3]. In order to claim that the w coordinate furnishes a complex structure, we need to examine the transition function between overlapping patches. Suppose we have

$$ds^2 = \rho_1(w_1, \bar{w}_1) |dw_1|^2 = \rho_2(w_2, \bar{w}_2) |dw_2|^2, \quad w_2 = F(w_1, \bar{w}_1) \quad (15)$$

By definition, we have

$$\begin{aligned} dw_2 d\bar{w}_2 &= (\partial_{w_1} F \overline{\partial_{w_1} F} + \overline{\partial_{\bar{w}_1} F} \partial_{\bar{w}_1} F) dw_1 d\bar{w}_1 + \partial_{w_1} F \overline{\partial_{\bar{w}_1} F} dw_1^2 + h.c. \\ &\propto dw_1 d\bar{w}_1 \end{aligned} \quad (16)$$

To preserve the orientation, we must have $\partial_{\bar{w}_1} F = 0$, i.e., the transition function is holomorphic. Therefore, we show that each metric naturally induces a unique complex structure that only depends on the conformal structure, as we can see from the Beltrami equation (13).

To complete the equivalence, we now show that a complex structure also naturally induces a metric that is unique up to conformal equivalence, i.e., a unique conformal structure. It suffices to analyzing two overlapping patches (U_1, z_1) and (U_2, z_2) in the given complex structure. On each patch, we first assign $|dz_1|^2$ and $|dz_2|^2$ as the trial metric. On the overlapping region $U_1 \cap U_2$, the two trial metrics differs by a rescaling

$$|dz_2|^2 = |\partial_{z_1} z_2|^2 |dz_1|^2$$

The next step is to smoothly interpolate $|dz_1|^2$ and $|dz_2|^2$ to define a metric globally. Formally, we use the partition of unity and write

$$ds^2 = \sum_{\alpha} \varphi_{\alpha}(p) |dz_{\alpha}(p)|^2, \quad \sum_{\alpha} \varphi_{\alpha}(p) = 1 \quad (17)$$

where α labels different patches and the smooth function $\varphi_{\alpha}(p) \neq 0$ only for $p \in U_{\alpha}$. For example, if only U_1 and U_2 overlaps in a small region, the metric reads

$$ds^2 = (\varphi_1(p) + \varphi_2(p) |\partial_{z_1} z_2|^2) |dz_1|^2$$

Different choices of φ_{α} or performing holomorphic coordinate transformation on each patch end up with metrics that are all conformally equivalent to each other. Therefore, each complex structure uniquely induces a conformal structure. Below, we will use the two words complex structure and conformal structure synonymously.

Conformal = hyperbolic Consider two conformally equivalent metric $g_{\mu\nu}(x) = e^{\sigma(x)}\widehat{g}_{\mu\nu}(x)$, then their curvatures are related by

$$R(x) = e^{-\sigma(x)}(\widehat{R}(x) - \Delta_{\widehat{g}}\sigma) \quad (18)$$

Therefore, the hyperbolic structure is the representative of the conformally equivalent metric that has a constant negative curvature $R(x) = -2$. Of course, it is not always possible to realize a constant negative curvature on every surface. The equivalence between conformal and hyperbolic structure holds only for surfaces with negative Euler characteristic.

An important result on two-dimensional surfaces is the uniformization theorem. Our discussion so far can be briefly summarized as the equivalence between the following two versions of the uniformization theorem:

Theorem 2.1 (Uniformization theorem).

Version 1: Every simply connected Riemann surface is biholomorphically equivalent to exactly one of the three Riemann surfaces: the Riemann sphere $\widehat{\mathbb{C}}$, the complex plane $\mathbb{C}$, and the unit disk $\mathbb{D}$.

Version 2: For a closed, oriented smooth surface S equipped with a conformal structure, there exists a Riemannian metric in the conformal class whose Gaussian curvature is constant K . If $g(S) = 0$, $K = 1$; if $g(S) = 1$, $K = 0$; if $g(S) \geq 2$, $K = -1$.

3 Space of hyperbolic surfaces

In this section, we discuss the space of hyperbolic surfaces, the so-called Teichmüller space. We first give its definition and discuss its distinction from the moduli space, another commonly used notion. We then explain the pair-of-pants decomposition. The pair-of-pants decomposition of a topological surface is far from being unique. We will see that how the geodesics under a hyperbolic metric help us to make a canonical choice (although still not unique). In some sense, hyperbolic spaces are fairly “rigid” and “compact”.

3.1 Teichmüller space

Let S be a compact surface with finitely many (perhaps zero) points removed from the interior. The Teichmüller space $\mathcal{T}(S)$ of S is the set of equivalence classes of marked hyperbolic structures (X, ϕ) on S :

$$\mathcal{T}(S) := \{(X, \phi)\}/\text{homotopy} \quad (19)$$

Note that S is equipped only with topology in this expression. Viewing the hyperbolic structure as a hyperbolic metric, we can also define the Teichmüller space as

$$\mathcal{T}(S) := \text{HyperMet}(S)/\text{Diff}_0(S) \quad (20)$$

Strictly speaking S is equipped with a coordinate system and a Riemannian metric here.

As is discussed in Sec. 2, we can also define the Teichmüller space as the space of inequivalent complex/conformal structures. It follows immediately that the Beltrami differential is a useful tool to describe the Teichmüller space. Let us first specify a reference complex/conformal structure via choosing a set of isothermal coordinates on every patch

$$U_\alpha : \widehat{ds^2} = e^{\sigma_\alpha(z_\alpha, \bar{z}_\alpha)} |dz_\alpha|^2 \quad (21)$$

This is our arbitrarily chosen starting point of the Teichmüller space. To explore the whole space, we add a Beltrami differential to deform the complex structure as follows

$$\widehat{ds^2} \rightarrow ds^2 = e^{\sigma'_\alpha(z_\alpha, \bar{z}_\alpha)} |dz_\alpha + \mu_\alpha(z_\alpha, \bar{z}_\alpha) d\bar{z}_\alpha|^2 \quad (22)$$

where the differential on the overlapping region should obey the $(-1, 1)$ transformation rule

$$\mu_\alpha(z_\alpha, \bar{z}_\alpha) = \left(\frac{\partial z_\beta}{\partial z_\alpha}\right)^{-1} \left(\frac{\partial \bar{z}_\beta}{\partial \bar{z}_\alpha}\right) \mu_\beta(z_\beta, \bar{z}_\beta) \quad (23)$$

Accordingly, we modify the prefactor from σ_α to σ'_α to guarantee ds^2 is single-valued in the overlapping region. Not every Beltrami differential gives rise to a distinct complex structure, many of them are equivalent to a coordinate transformation. Specifically, the following deformation on each patch

$$U_\alpha : \quad \mu_\alpha = (\partial z_\alpha / \partial \bar{\zeta}_\alpha) (\partial z_\alpha / \partial \zeta_\alpha)^{-1}$$

is equivalent to a coordinate transformation $z_\alpha = z_\alpha(\zeta_\alpha, \bar{\zeta}_\alpha)$. Let $\mathcal{B}(S)$ denote the space of Beltrami differential, whose absolute value is bounded from above by 1 everywhere. The Teichmüller space can be defined by

$$\mathcal{T}(S) = \mathcal{B}(S) / \text{Diff}_0(S) \quad (24)$$

Despite the fact that $\mathcal{B}(S)$ is an infinite dimensional space, we will see that the Teichmüller space, remarkably, has finite dimensions.

Teichmüller space vs moduli space Another commonly used concept is the moduli space, defined as

$$\mathcal{M}(S) = \text{HyperMet}(S) / \text{Diff}(S) \quad (25)$$

Namely, we further mod out diffeomorphisms that are not connected to the identity. The extra diffeomorphisms are also called the mapping class groups, i.e.,

$$\text{MCG}(S) = \text{Diff}(S) / \text{Diff}_0(S) \quad (26)$$

Therefore, by definition, we have

$$\mathcal{M}(S) = \mathcal{T}(S) / \text{MCG}(S) \quad (27)$$

For example the Teichmüller space of a pair of pants is isomorphic to $\mathbb{R}_+^3$, but the moduli space, obtained by modding out permutation, is not.

3.2 Pair-of-pants decomposition

Owing to the equivalence established in Sec. 2, it is natural to study hyperbolic structure from a local perspective. This motivates us to first find the minimal building block of a compact surface, and conquer them separately.

Let S be a compact surface with $\chi(S) < 0$. A pair-of-pants (POPs) decomposition is a collection of disjoint simple closed curves in S with the property that when we cut S along these curves we obtain a disjoint union of POPs. A few comments are in order:

- Since any closed curve on a single POP is homotopic either to a single point or to a boundary component, the chosen curves in this collection are not homotopic to each other.
- A single POP has $\chi(P) = -1$. The resulting surface of gluing two POPs still has $\chi = -1$. As a result, a POP decomposition of a surface S yields $-\chi(S)$ POPs.
- For a compact surface without any boundary or punctures, a POP decomposition requires $3g - 3$ closed curves and yields $2g - 2$ POPs.

For a topological surface, the choice of these closed curves for a POP decomposition has a huge redundancy. On a hyperbolic surface, closed geodesics offer us a finite number of canonical choices.

3.3 Geodesics on hyperbolic space

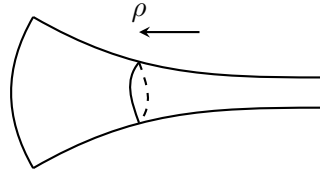
A POP decomposition requires a maximal collection of closed curves that are not homotopic to each other. Within the homotopic class of a given closed curve, there are still infinitely many choices. Fortunately, the geodesic is unique, which offers us a canonical choice. More precisely, we have

Theorem 3.1. *Let S be a compact hyperbolic surface and let c be a homotopically non-trivial closed curve. Then the following hold:*

1. c is freely homotopic to a unique closed geodesic γ (there exists a continuous map to interpolate the two closed curves)
2. γ is either contained in ∂S or $\gamma \cap \partial S = \emptyset$
3. If c is simple (no self-intersection), then γ is simple

We refer the reader to Ref. [1] for the proof and just gives two comments to help develop some intuition:

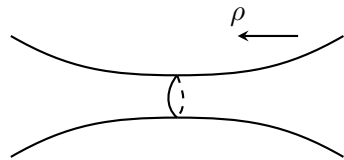
- Intuitively, one can deform a closed curve and identify the geodesic as the saddle point of this deformation (I just translate the variational principle into words). Since a hyperbolic surface locally looks like a saddle surface, small deformation of a geodesic cannot keep its length invariant, implying the uniqueness.
- The compactness of the surface is important for the existence of the geodesic γ . For example, consider a cusp, with a longitudinal coordinate $\rho \in (-\infty, 0]$ and an angular coordinate $\tau \in [0, 2\pi]$, and a hyperbolic metric



$$ds^2 = d\rho^2 + e^{2\rho} d\tau^2 \quad (28)$$

For a closed curve traversing the angular direction (draw above), we can keep decreasing its length by pushing it towards $\rho \rightarrow -\infty$. Therefore, it is not homotopic to any geodesic.

- On a compact surface, we expect the closed geodesics to correspond to the thinnest necks of the handles. In the neighborhood of the geodesics, it is convenient to describe the hyperbolic surface via the Fermi coordinate



$$ds^2 = d\rho^2 + \cosh^2 \rho d\tau^2 \quad (29)$$

This is called the hyperbolic collar. The closed geodesic is the curve at $\rho = 0$. Other geodesics are the $\tau = \text{const}$ curves.

By construction, decomposing a hyperbolic surface with totally geodesic boundaries along a collection of disjoint geodesics gives another hyperbolic surface with totally geodesic boundaries. Therefore, this procedure yields surfaces with smaller complexity, and an inductive method for understanding the hyperbolic structure of the original surface. Below, we will use geodesics to perform a POP decomposition.

Theorem 3.2. *Let S be a compact hyperbolic surface of genus $g \geq 2$, there exist simple closed geodesics $\gamma_1, \gamma_2, \dots, \gamma_{3g-3}$ that are pairwise disjoint. They together decomposes S into pairs of pants.*

In our following discussion, all the boundaries of a POP are geodesics.

We close this section with a fun fact about geodesics on hyperbolic surfaces that is not directly related to our discussion of POP decomposition

Theorem 3.3. *Let S be a compact hyperbolic surface. Given $L > 0$, there are finitely many closed geodesic on S that have length less than L .*

This is in stark contrast to the situation on a flat torus or a sphere. The closed geodesics on a compact hyperbolic surface are, in some sense, the landmarks, that are “rigid” and discrete.

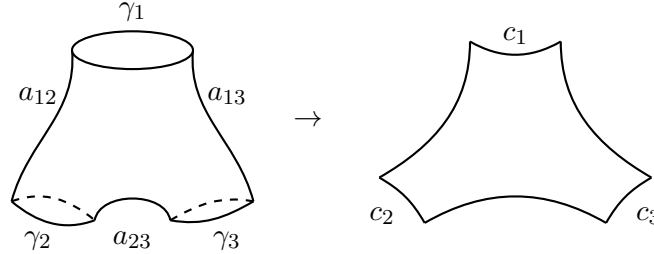
4 A single pair of pants: length parameter

In this section, we examine a single pair of pants and show that its hyperbolic structure is uniquely determined by the lengths of its three boundaries. We will show $\mathcal{T}(P) = \mathbb{R}_+^3$. We then discuss the regime of vanishing boundary lengths.

4.1 Teichmüller space of a pair of pants

The strategy is to show that a hyperbolic POP is in one-to-one correspondence to a right-angled geodesic hexagon, which is uniquely determined by the three lengths of its alternative edges.

Given a POP depicted below, every two boundaries are connected by a unique shortest geodesic that is also perpendicular to the boundaries, denoted by a_{12}, a_{23}, a_{13} . These six geodesics γ_i and c_{ij} further cut the POP into two so-called right-angled geodesic hexagons, drawn on the right.



We now quote a result from the hyperbolic trigonometry:

Theorem 4.1. *Given three positive numbers x, y, z , there is a unique right-angled geodesic hexagon $\alpha, a, \beta, b, \gamma, c$ such that $\alpha = x, \beta = y, \gamma = z$. Namely, a geodesics hexagon is uniquely determined by the lengths of its alternating edges.*

Therefore the two hexagons are isometric. As a corollary, the end points of the perpendiculars cut each boundary of the POP into two arcs of the same length. This procedure is summarized in the following proposition:

Proposition 4.2. *Let P be an arbitrary POP. For every pair of boundary geodesics of P there exists a unique simple common perpendicular. The three perpendiculars together decompose S into two isometric right-angled geodesic hexagons.*

We refer the reader to Chapter 3 of Ref. [1] for a rigorous proof. Here the non-trivial part of this proof is in the uniqueness of the perpendiculars and the right-angled geodesic hexagon. Given these results, we have

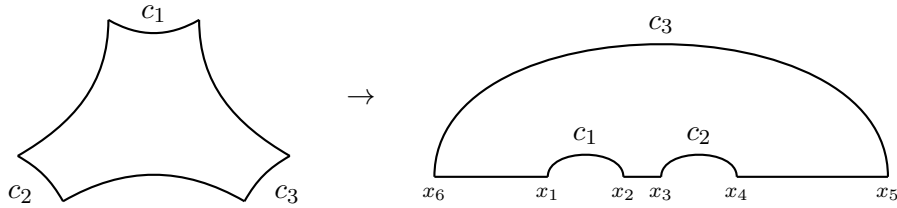
Theorem 4.3. *For any triple of positive real numbers ℓ_1, ℓ_2, ℓ_3 , there is a unique pair of pants P with boundary geodesics $\gamma_1, \gamma_2, \gamma_3$ of length ℓ_1, ℓ_2, ℓ_3 , respectively.*

To define the Teichmüller space, we need to remember that the pair of pants is a marked space. In particular, permutation among the three boundaries are not allowed when defining the equivalence relation. This way we see that the Teichmüller space is

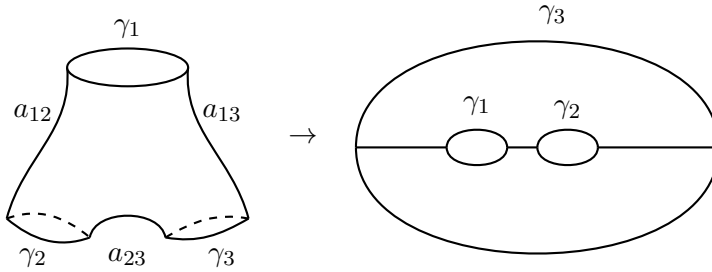
$$\mathcal{T}(P) = \mathbb{R}_+^3 \quad (30)$$

The three lengths constitute the so-called length parameter in the Fenchel-Nielsen coordinate.

It follows from this decomposition into two hexagons that we can also represent a pair of pants as the complex plane with three holes. Specifically, we recall the Riemann mapping theorem: *any non-empty simply connected open subset is biholomorphic to the open unit disk*. Therefore, the right-angled geodesic hexagon, a simply connected region inside the Poincaré disk, can be conformally mapped to a region on the upper half-plane drawn on the right:



The six vertices are on the real axis, the three half boundary geodesics are semi-circles and the perpendiculars are along the real axis. We can use an $SL(2, \mathbb{R})$ transformation to fix the centers of the semi-circles c_1, c_2 and c_3 . We are then left with the radius of the three semi-circles, corresponding to the three boundary lengths. This way we can represent the full POP by the complex plane with three holes centered at $z = 0, 1$ and ∞ :



In principle, the Riemann mapping theorem allows us to map the hexagon to an arbitrary simply connected region. This particular choice, mapping each boundary to a circle, has the advantage that the hyperbolic metric near the circle on the complex plane is smooth. We will provide an explicit form when discussing the vanishing boundary length limit.

4.2 Collar lemma and collar theorem

In this section, we provide more intuitions for how the boundary lengths constrain the geometry of a pair of pants, the so-called collar lemma. It will also offer a glimpse into a fundamental property of hyperbolic surfaces, known as the collar theorem, which says that the small closed geodesies have large tubular neighborhoods which are topological cylinders.

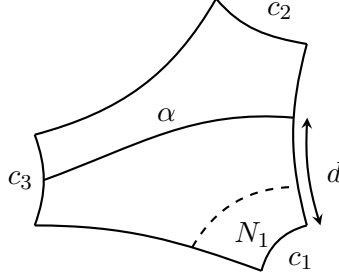
The collar lemma says the following (It might look differently from the textbook, e.g. Ref. [2]. This version is what the proof directly work on, and it assists our later discussion better)

Lemma 4.4 (Collar Lemma). *Consider a hyperbolic pair of pants P with three boundaries all being geodesics. Let γ be one boundary and $N_\gamma = \{x : d(x, \gamma) < w\}$ an annulus neighborhood. If we have*

$$\sinh w \sinh \frac{\ell(\gamma)}{2} = 1 \quad (31)$$

Then N_γ does not contain any of the other two boundary curves.

To prove this lemma, we divide P into two isometric right-angled hexagons H and H' . We draw H below and label the alternating edges that correspond to the boundaries of P by c_1, c_2 and c_3 . Without loss of generality, choose $c_1 = \gamma$ and $N_1 \subset N_\gamma$. Let α be the shortest geodesic from c_3 to the opposite side.



The right-angled pentagon formula says that if a and b are adjacent sides of a right-angled hyperbolic pentagon and, c is the side opposite their common vertex, then

$$\sinh a \sinh b = \cosh c \quad (32)$$

Applying this formula to the pentagon below α , we have

$$\sinh \ell(c_1) \sinh d = \cosh c \geq 1 \quad (33)$$

where $d = \text{dist}(c_1, \alpha)$, c is the length of c_3 below α . Therefore

$$\sinh d \geq \frac{1}{\sinh \ell(c_1)} \quad (34)$$

Thus N_{c_1} is strictly contained by the pentagon below α . Similarly N_{c_2} is contained by the pentagon above α . As a result $N_{c_1} \cap N_{c_2} = \emptyset$. By symmetry, we also know $N_{c_1} \cap N_{c_3} = N_{c_2} \cap N_{c_3} = \emptyset$.

Remark The width of the half collar is maximal in the following sense. Consider a sequence of pairs of pants, in which $\ell(\gamma_3) \rightarrow 0$ and $\ell(\gamma_1), \ell(\gamma_2)$ are fixed. In this case, we apply the hexagon formula in a different way and have

$$\sinh \text{dist}(c_i, \alpha) = \frac{\cosh \ell(c_3)/2}{\sinh \ell(c_i)} \rightarrow \frac{1}{\sinh \ell(c_i)}, \quad i = 1, 2 \quad (35)$$

In this case, increasing the width by little can violate the disjoint property in the lemma.

We can then extend this result to the entire hyperbolic surface S . Combining the collar lemma with Theorem. 3.2, we immediately have

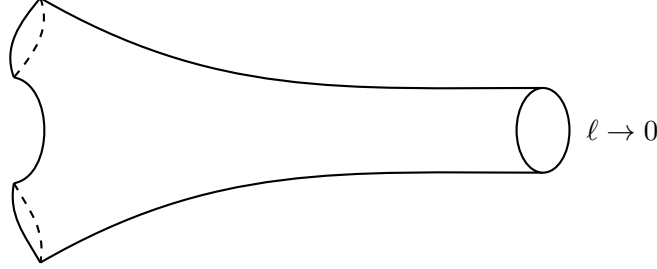
Theorem 4.5. *Let S be a compact hyperbolic surface with genus $g \geq 2$, and $\gamma_1, \dots, \gamma_{3g-3}$ be pairwise disjoint simple closed geodesics. Then the collars*

$$\mathcal{C}(\gamma_i) = \{x \in S : \text{dist}(x, \gamma_i) \leq w(\gamma_i)\}, \quad \sinh w(\gamma_i) \sinh \frac{\ell(\gamma_i)}{2} = 1$$

are disjoint from each other. Each collar is isometric to the cylinder $[-w(\gamma_i), w(\gamma_i)] \times S^1$, with the Fermi coordinate $\rho \in [-w(\gamma_i), w(\gamma_i)]$, $\tau \in [0, 1]$ and the hyperbolic metric $ds^2 = d\rho^2 + \ell(\gamma_i)^2 \cosh^2 \rho d\tau^2$.

4.3 Cusp limit

By the collar lemma, a boundary with a smaller length can have a wider half-collar. As $\ell \rightarrow 0$, the boundary geodesic is pushed to the infinity from the bulk of the pants, hence the name “cusp limit”, as is drawn below



In this section, we examine this cusp limit. We will relate the asymptotic behavior of the length parameter to our geometric intuitions, and also discuss what a cusp looks like on the complex plane.

Colloquially speaking, the half-collar of the pinched cuff becomes a “thin and long” neck near the cusp limit. More precisely, we embed the pair of pants inside the three-dimensional Euclidean space, where the half-collar looks “thin and long”. Quantitatively, the embedded surface near the pinched cuff can be well approximated by a cylinder with length L and circumference $\varepsilon \ll L$. We use $u \in [0, L]$ for the longitudinal coordinate, $\theta \in [0, \varepsilon]$ for the transverse coordinate, and write the metric as $ds^2 = du^2 + d\theta^2$. In order to compute the length parameter, we need to treat the cylinder as part of a hyperbolic space. Here it is convenient to use the Fermi coordinate (ρ, τ) with $\rho \in [0, w]$ and $\tau \in [0, \ell]$ and write the hyperbolic metric as $ds^2 = d\rho^2 + \cosh^2 \rho d\tau^2$. Namely we map the boundary $u = 0$ to $\rho = 0$, and the other side $u = L$ to a large w that obeys the collar lemma. Since the two Euclidean and hyperbolic metrics are conformally equivalent, setting $\rho = \rho(u)$, $\tau = \ell\theta/\varepsilon$, we have

$$d\rho^2 + \cosh^2 \rho d\tau^2 = \left(\frac{d\rho}{du}\right)^2 du^2 + \cosh^2 \rho \left(\frac{\ell}{\varepsilon}\right)^2 d\theta^2 = \lambda(u)^2 (du^2 + d\theta^2) \quad (36)$$

Thus we have

$$\lambda(u) = \frac{d\rho}{du} = \frac{\ell}{\varepsilon} \cosh \rho \quad (37)$$

Solving this differential equation and using the fact that $w \gg 1$, we have

$$\ell \frac{L}{\varepsilon} = \int_0^w \frac{d\rho}{\cosh \rho} = \frac{\pi}{2} - \cot^{-1}(\sinh w) \approx \frac{\pi}{2} - \frac{1}{\sinh w} \quad (38)$$

We can use the collar lemma to estimate that $1/\sinh w \leq \sinh \frac{\ell}{2} \sim \ell$, which can be ignored in the above equation. Therefore, we have

$$\ell = \frac{\pi}{2} \frac{\varepsilon}{L} + \mathcal{O}((\varepsilon/L)^2) \quad (39)$$

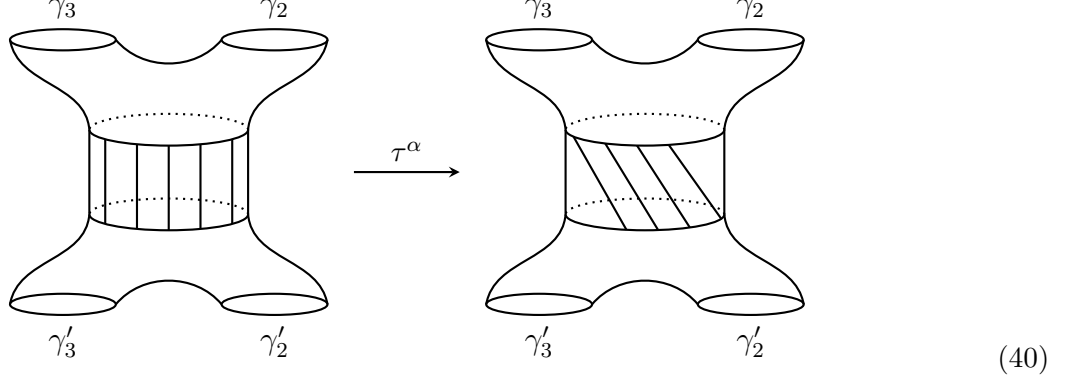
We can also represent the half-collar on the complex plane. We choose to map it to an annulus centered at the origin $z = 0$, the boundary at $u = L$ to the outer circle at $|z| = r_0$, and the geodesic boundary at $u = 0$ to the inner circle with radius

$$R/r_0 \approx e^{-\frac{2\pi L}{\varepsilon}} = e^{-\pi^2/\ell}$$

Note that we need to choose a small and finite r_0 so that we can map one of the other cuff to a hole centered at $z = 1$. In the cusp limit $\ell \rightarrow 0$, the hole on the complex plane vanishes.

5 Gluing two pairs of pants: twist parameter

Having understood a single pair of pants, let us now try to glue two pants together. Take two pants P and P' with $\ell = \ell(\gamma_1) = \ell(\gamma'_1)$, then we can glue them together along this boundary as is drawn below on the left-hand side



The resulting surface is often called an X -piece for obvious reasons. On each side of the glued boundary γ_1 , the surface locally is a half-collar, characterized by the Fermi coordinate. Therefore, we can twist one side during the gluing process and the resulting surface is still completely smooth near the glued region. This naturally leads us to the twist parameter.

Specifically, we start with the un-twisted X -piece. The collar neighborhood of γ_1 is described by the Fermi coordinate and the metric

$$ds^2 = d\rho^2 + \ell^2 \cosh^2 \rho d\tau^2, \quad \tau \in [0, 1]$$

where we let $\rho < 0$ ($\rho > 0$) belong to P (P'). The $\tau = \text{const}$ curves are all geodesics perpendicular to γ_1 , which traverse smoothly from P to P' . To define the twisted X -piece, let $\gamma_1(\tau)$ and $\gamma'_1(\tau)$ denote the boundary geodesics parameterized by τ . The gluing with a twist parameter α is given by

$$\gamma_1(\tau) = \gamma'_1(\tau + \alpha/2\pi), \quad \alpha \in \mathbb{R} \quad (41)$$

It is confusing but important to notice that the twist parameter α is treated as a real number instead of a periodic parameter. For example, if you compare the two surfaces with $\alpha = 0$ and $\alpha = 2\pi$ by tracing geodesics, they look identical. However, they are regarded as two different elements in the Teichmüller space (see the figure in Eq.(6)). To see why the twist leads to a different hyperbolic structure, first recall that there is a unique orthogeodesic between any two boundaries of a hyperbolic surface. As we twist the gluing, the shortest arc connecting two boundary components clearly changes. Since the geodesic length is an invariant under diffeomorphism, it implies that the twist yields an inequivalent hyperbolic structure.

Based the discussion so far, here is a simple method that determines the twist modulo 2π :

1. Choose one boundary from each pair of pants, e.g., γ_3 and γ'_2 .
2. Draw an arc that connects γ_3 and γ'_2 . There is no canonical choice and the arc can even wind around the waist of the X -piece.
3. Homotope the arc in each pair of pant separately to the shortest geodesic that connects γ_3 and γ_1 , γ'_2 and γ_1 , respectively. This is the shortest geodesic connecting γ_3 and γ'_2 .
4. Compute the amount $\Delta\tau$ that deformed arc traverses along γ_1 .

5. The twist parameter θ is given by

$$\theta = 2\pi\Delta\tau \quad (42)$$

We use this method in Ref. [4].

6 Fenchel-Nielsen coordinate

Our discussion so far can be summarized by the Fenchel-Nielsen coordinate. For an arbitrary closed surface S of genus $g \geq 2$, we can choose $3g - 3$ disjoint closed geodesics and divide S into pairs of pants. The hyperbolic structure of the original surface is then parameterized by $3g - 3$ length parameters $\ell_1, \dots, \ell_{3g-3}$ that are the lengths of the chosen geodesics, and another $3g - 3$ twist parameters $\theta_1, \dots, \theta_{3g-3}$ that describe the twist angles for gluing these pairs of pants. These are called the Fenchel-Nielsen coordinate of the Teichmüller space. We have

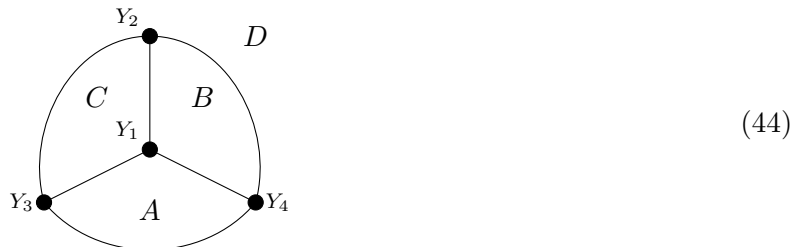
$$\begin{aligned} \text{FN} : \mathcal{T}(S) &\rightarrow \mathbb{R}_+^{3g-3} \times \mathbb{R}^{3g-3} \\ (X, f) &\mapsto (\ell_1, \dots, \ell_{3g-3}, \theta_1, \dots, \theta_{3g-3}) \end{aligned} \quad (43)$$

Given the choice of closed geodesics, the length parameters are automatically determined. To solve for the twist parameters, we use the following method, which, again fixes twists modulo 2π :

1. Fix a choice of the POP decomposition (often not unique)
2. Choose a collection of disjoint simple closed curves $\beta_1, \dots, \beta_n$, called seams. On each pair of pant P , these seams should exactly give three disjoint arcs connecting the boundary curves of P pairwise.
3. Follow the prescription in Sec. 5 to calculate the twist.

The choice of seams is not unique. But sometime there are canonical choices. For example, if the surface S is constructed through cutting and gluing simpler, untwisted surfaces, then we can first specify a set of seams on the untwisted surface and use them to infer an educated choice on S . Below is a concrete example to demonstrate how it might work in practice.⁴

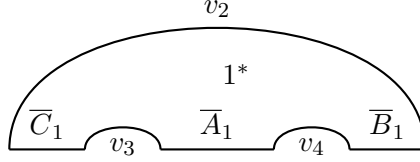
Example: twist parameters of a glued surface We start with a genus-3 surface, topologically equivalent to the surface of a normal steering wheel, whose combinatorial pattern is shown below (see Ref. [1] for more usage of the combinatorial pattern)



where $Y_{i=1,\dots,4}$ label the four pairs of pants, A, B, C, D label the perpendiculars on each pair of pants. For example A_1 means the orthogeodesic, or perpendicular, on Y_1 connecting the 13 boundary and 14 boundary. We assume this surface has no twist. Namely, this surface is reflection symmetric with respect to the paper so that all orthogeodesics lie in the paper plane. Now, we take three copies of this base surface and generate a new one via the following cut and paste procedure

⁴The example is use for the calculation of Rényi modular commutator in Ref. [4], i.e., the $n = 1$ case briefly analyzed in Appendix E.

1. Cut the base surface into four pairs of pants $Y_1, \dots, Y_4$ as shown above
2. Cut each pair of pants into two geodesic hexagons by the perpendiculars. Denote the perpendiculars in the two hexagons by, e.g., A_i and $\bar{A}_i$ etc. For example, we represent one half of the cut Y_1 on the complex plane by

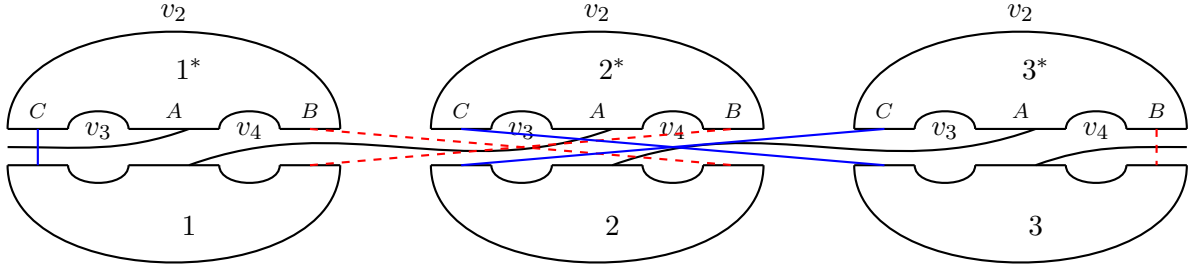


Here v_2, v_3, v_4 denotes which pants it connects to.

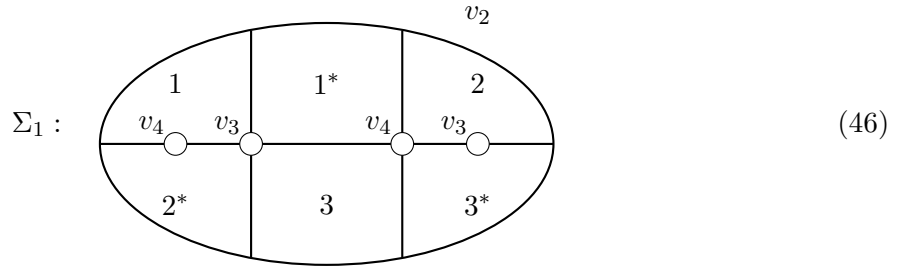
3. Glue pairs of pants across replicas along the perpendiculars. Take Y_1 as a concrete example. There are three copies $Y_1^{(\alpha=1,2,3)}$. We glue A_1^α to $\bar{A}_1^{\pi_A(\alpha)}$, B_1^α to $\bar{B}_1^{\pi_B(\alpha)}$ and C_1^α to $\bar{C}_1^{\pi_C(\alpha)}$. Here, the permutations are

$$\pi_A = (1, 2, 3), \quad \pi_B = (1, 2)(3), \quad \pi_C = (1)(2, 3), \quad \pi_D = (1)(2)(3) \quad (45)$$

This procedure yields $\Sigma_{i=1,\dots,4}$, each of which is topologically a sphere with five holes. We draw the gluing procedure for three Y_1 's below, where the black, red and blue lines are for A, B, C perpendiculars respectively



It is easier to represent the resulting surface Σ_1 on the complex plane by



Here the infinity connects to Σ_2 . Viewing Σ_1 as a covering surface of Y_1 , then each hole on Y_1 can give rise to multiple holes on Σ_1 with non-trivial ramifications. Specifically, for the hole v_i adjacent to the perpendiculars I and J , we define the relative permutation $\pi_{IJ} = \pi_I \pi_J^{-1}$. Then the number of resulting holes is the number of cycles in π_{IJ} , and the ramification indices equal to the lengths of the cycles. For example, consider the hole v_3 that is adjacent to the A and C perpendiculars. We have $\pi_{AC} = \pi_A \pi_C^{-1} = (1, 2)(3)$. Indeed, there are two v_3 holes on the resulting surface, one of which has a ramification index 1 and the other has 2.

Similarly, we have

$$\Sigma_2 : \begin{array}{c} \text{Diagram of surface } \Sigma_2 \text{ with holes } v_1, v_3, v_4 \text{ and regions } 1, 1^*, 2, 2^*, 3, 3^* \end{array} \quad (47)$$

$$\Sigma_3 : \begin{array}{c} \text{Diagram of surface } \Sigma_3 \text{ with holes } v_1, v_2, v_4 \text{ and regions } 1, 1^*, 2, 2^*, 3, 3^* \end{array} \quad (48)$$

$$\Sigma_4 : \begin{array}{c} \text{Diagram of surface } \Sigma_4 \text{ with holes } v_1, v_2, v_3 \text{ and regions } 1, 1^*, 2, 2^*, 3, 3^* \end{array} \quad (49)$$

Each surface Σ_i has five holes, two of them have a unit ramification index.

Note that all four resulting surface have holes centered on the real axis. Moreover, the orthogeodesics connecting every two boundary curves lie along the real axis. This is the consequence of a local reflection symmetry. Namely, given a vertex v and its three perpendiculars I, J and K , we define $\pi_{IJ} = \pi_I \pi_J^{-1}$ and $\pi_{JK} = \pi_J \pi_K^{-1}$. Having local reflection symmetry means that there exists a v -dependent permutation π_v satisfying

$$\pi_v^2 = 1, \quad \pi_v \pi_{IJ} \pi_v = \pi_{IJ}^{-1}, \quad \pi_v \pi_{JK} \pi_v = \pi_{JK}^{-1}. \quad (50)$$

4. Glue Σ_i accordingly to get the new surface Σ . The combinatorial pattern reads

$$\begin{array}{c} \text{Diagram of the resulting surface } \Sigma \text{ with vertices } Y_1, Y_2, Y_3, Y_4 \text{ and arcs (solid black and dashed red)} \end{array} \quad (51)$$

where the dashed red arcs denote the handles obtained from gluing unit-ramified holes, and black arcs are ones with higher ramifications. The resulting surface has a genus $g = 7!$

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